

This assignment will not be graded and is only for practice.

Level 1:

1) Consider a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as:

$$f(x, y) = \begin{cases} x^2 \sin \frac{1}{x} + y & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

Statement 1: f_y is continuous at the origin.

Statement 2: f_x is continuous at the origin.

Enter the number of correct statements.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

2) A function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ satisfies that $f_x(x, y) = c$ and $f_y(x, y) = k$ of all points of $\mathbb{R}^2$ **1 point**, where c and k are some constants, then which of the following option(s) is (are) true?

- ☐ The tangent line exists at any point of $\mathbb{R}^2$ in any direction.
- ☐ The tangent lines at the origin do not lie on a plane.
- ☐ There exist points in $\mathbb{R}^2$ for which tangent points in all directions do not exist.
- ☐ $f(x, y)$ can be $f(x, y) = cx + ky + \alpha$, where α is any real number.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The tangent line exists at any point of $\mathbb{R}^2$ in any direction.

$f(x, y)$ can be $f(x, y) = cx + ky + \alpha$, where α is any real number.

3) Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and suppose partial derivatives of $f(x, y)$ with respect to x and y are continuous at a point (a, b) . Then which of the following option(s) is (are) true? **1 point**

- ☐ $f_x(a, b) = \lim_{(x, y) \rightarrow (a, b)} f_x(x, y)$

☐ $f_y(a, b) = \lim_{(x,y) \rightarrow (a,b)} f_y(x, y)$

☐ $f_x(a, b) = \lim_{(x,y) \rightarrow (a,b)} f_x(x, y).$

☐ $f_y(a, b) = \lim_{(x,y) \rightarrow (a,b)} f_y(x, y)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f_x(a, b) = \lim_{(x,y) \rightarrow (a,b)} f_x(x, y)$$

$$f_y(a, b) = \lim_{(x,y) \rightarrow (a,b)} f_y(x, y)$$

4) Consider a function $f(x, y) = x^2 + y^2$. Which of following is the tangent at the point **1 point** (1,2) in the direction of (1, 1)?

☐ $x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$

☐ $x = 1 + t, y(t) = 2 + t, z(t) = 5 + 6t$

☐ $x = 1, y(t) = 2 + t, z(t) = 5 + \frac{6t}{\sqrt{2}}$

☐ $x = 1 + t, y(t) = 2 + t, z(t) = 4 + \frac{6t}{\sqrt{2}}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$$

Level 2:

5) Consider a function

1 point

$$f(x, y) = \begin{cases} \frac{x}{y} & \text{if } y \neq 0 \\ 0 & \text{if } y = 0. \end{cases}$$

Which of the following is the tangent line passing through the point (1, -1) and parallel to the line $x(t) = \frac{t}{\sqrt{5}}, y(t) = 1 + \frac{2t}{\sqrt{5}}, z(t) = -\frac{3t}{\sqrt{5}}$?

- ☐ $x(t) = 1 + t, y(t) = -1 + 2t, z(t) = -\frac{3}{\sqrt{5}}t$
- ☐ $x(t) = 1 + t, y(t) = 1 + 2t, z(t) = -1 - \frac{3}{\sqrt{5}}t$
- ☐ $x(t) = 1 + \frac{t}{\sqrt{5}}, y(t) = -1 + \frac{2t}{\sqrt{5}}, z(t) = -1 - \frac{3t}{\sqrt{5}}$
- ☐ $x(t) = -1 + t, y(t) = -1 + 2t, z(t) = -1 - \frac{3}{\sqrt{5}}t$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$x(t) = 1 + \frac{t}{\sqrt{5}}, y(t) = -1 + \frac{2t}{\sqrt{5}}, z(t) = -1 - \frac{3t}{\sqrt{5}}$$

6) Consider a function $f(x, y, z) = 2x^2 + xy + z^2$. Which of following is the tangent at the point (1,1,1) in the direction of (1, 0, 0) **1 point**

- ☐ (1, 1, 1, 4 + 5t)
- ☐ (1 + t, 1, 1, 4 + t)
- ☐ (1, 1 + t, 1 + t, 4 + 5t)
- ☐ (1 + t, 1, 1, 4 + 5t)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$(1 + t, 1, 1, 4 + 5t)$$

7) Consider the following functions f defined as:

1 point

$$f(x, y, z) = xyz.$$

Then the tangent line at the point $(1,0,1)$ in the direction of $(1,1,1)$ is parallel to the line

- ☐ $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$
- ☐ $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = t\sqrt{3}$
- ☐ $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = 1 + \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$$

Your score is: 0/7